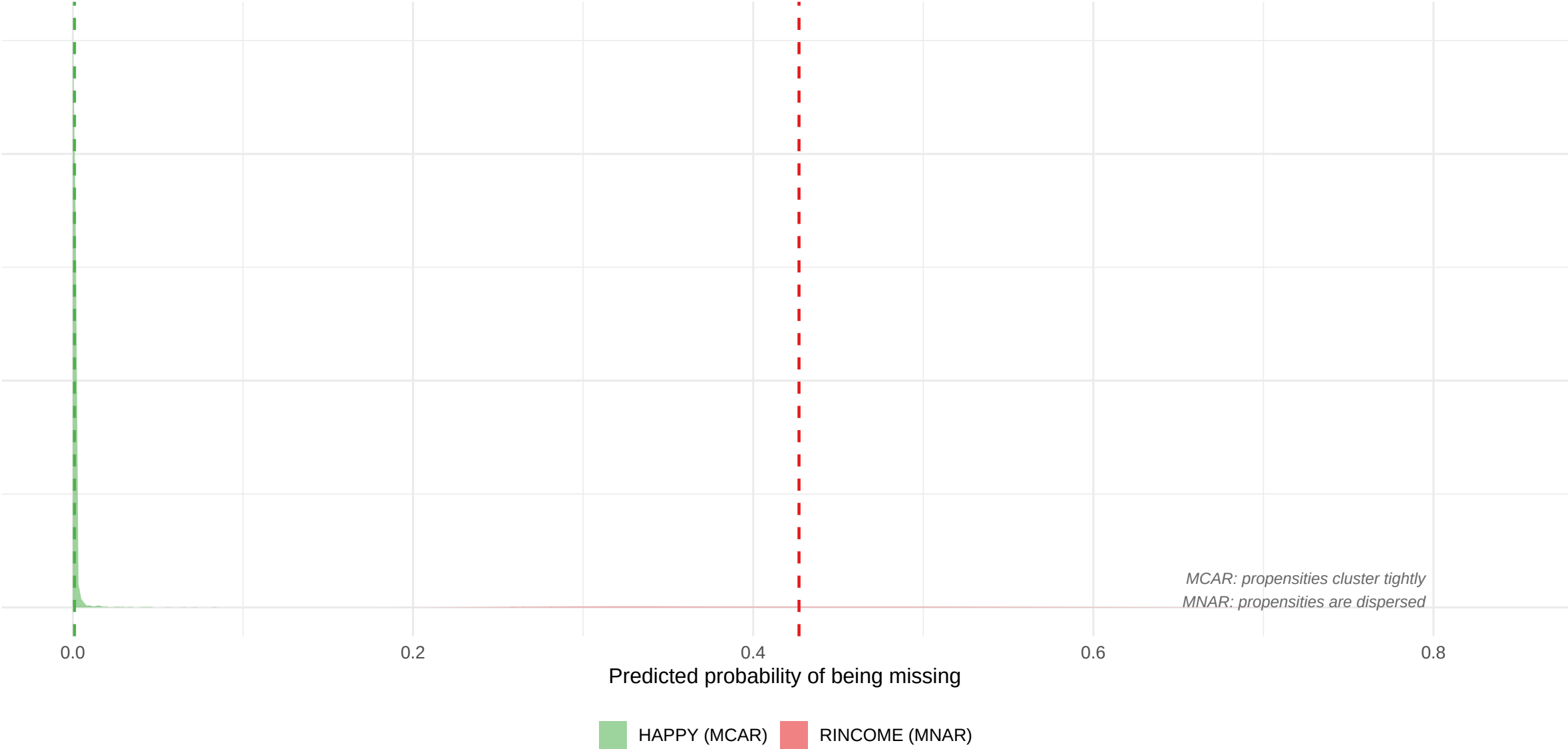


Advanced GSS Analysis: Propensity Diagnostics & Missing Data Strategies

General Social Survey 2018 (n = 2,348). Design: svydesign(id=~vpsu, strata=~vstrat, weights=~wtssall, nest=TRUE).
survey.lonely.psu = 'adjust'. All models use svyglm() with quasibinomial().

A. Propensity to Be Missing: MCAR vs MNAR

Design-weighted logistic regression (svyglm, quasibinomial) predicting missingness from age + education.
Tight clustering = MCAR (demographics don't predict missingness). Wide spread = MNAR (demographics predict who is missing).



B. Missing Income Strategy: Impact on Education Estimates

Mean years of education (GSS 2018) under four missingness strategies.
Conditioning on MNAR income missingness biases education estimates. IPW partially corrects.

